

**University of Calcutta**  
**Mode of Examination: Online**  
**M.Sc. Semester – I Examination, 2021**  
**Subject: Computer Science**  
**Paper Code & Name: CSMC102 (Data Structures and Algorithms)**  
**Full Marks: 70**

**Date: 05.03.2022**

**Time and Duration: 12.00 PM – 3:00 PM (3:00 Hours)**

**Please note the following instructions carefully:**

- **Promise not to commit any academic dishonesty.**
  - **Marks will be deducted if the same/similar answers are found in different answer scripts.**
  - **Candidates are required to answer in their own words as far as applicable.**
  - **The figures in the margin indicate full marks.**
  - **Each page of the answer-scripts should have your University Roll # on the right-top corner.**
  - **The name of the scanned copy of the answer script will be of the following format:  
CSMC102-DSA-University-Roll-Number.pdf  
(Example: CSMC102-DSA-C91-CSC-2110xy.pdf)**
  - **The subject of the mail should be the file name only.**
  - **The scanned answer script is to be sent to cucse2020@gmail.com**
  - **The file should have the top page (Page #1) as an index page, mentioning all page numbers against the answer of each question number.**
  - **The answer script may not be accepted after the scheduled time.**
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Answer **Question Nos. 1 and 2**, and **any four** from the rest.

**1. Answer any five from the following:** **[2×5 = 10]**

- (a) Clearly state and show the steps to construct an equivalent binary tree of the following tree: (J(K(L,M), N(O), P(Q(R), S(T,U), V))).
- (b) Instead of single rotation, why is it required to employ double rotation to make a binary tree height-balanced?
- (c) Relatively compare B-tree and B<sup>\*</sup>-tree.
- (d) Suppose that you are given three different weights, X, Y, and Z. Draw a decision tree to compare the weights in descending order.
- (e) Show that the external path length of a 2-tree is minimum when all the tree leaves are on the same level or on two adjacent levels.

- (f) Suppose a problem with  $n$  inputs is solved by a divide-and-conquer algorithm that partitions the problem into four equal parts, and the solutions of the subproblems are combined in  $O(n^2)$  steps. Write the recurrence relation and give an asymptotic upper bound.
- (g) Given a flow network, give steps to find the minimum cut.
- (h) State true or false with justification: Divide-and-conquer algorithm to multiply two large integers takes  $O(n^{1.59})$  time in the worst case.

**2. Answer any five from the following:**

**[4×5 = 20]**

- (a) Define the following notations and state how they are related: Big-Oh, Big-Omega, and Big-Theta.
- (b) Obtain the addressing function for the element  $A[s_1, s_2, \dots, s_n]$  in an array declared as  $A[l_1 .. u_1, l_2 .. u_2, \dots, l_n .. u_n]$ . Assume a row-major representation of the array with one word per element and  $\alpha$  the address of  $A[l_1, l_2, \dots, l_n]$ . Show that the function is linear.
- (c) What do you mean by graph traversal? Devise an algorithm to identify the presence of a cycle in a graph. Consider a suitable graph instance and show how the algorithm works.
- (d) Define and give an example of binomial heap. Prove that a binomial heap contains at most one binomial tree whose root has a given degree.
- (e) Consider the problem of incrementing  $n$ -digit binary integers and show that its actual cost is  $O(n)$ ; however, the amortized cost is independent of  $n$ .
- (f) Relatively compare NP, NP-complete, and NP-hard classes of problems with examples.
- (g) You are given a set of integers  $A = \{a_1, a_2, \dots, a_n\}$ , an integer  $k \in A$ , and  $p$ , where  $1 \leq p \leq n$ . Give an  $O(n)$  time algorithm to report if  $k$  is the  $p$ -th smallest element in the array. Justify the time complexity.
- (h) Find optimal prefix code for the characters  $a, b, c, d$ , and  $e$ , whose frequencies are respectively 32, 7, 5, 35, and 20. Show the Huffman tree and the code for each character.

- 3. (a)** Define heap. Consider the following data elements, and illustrate only the important passes to show how *HeapSort* works.

{21, 19, 4, 43, 28, 17, 1, 36, 11, 14, 46, 25}

**[1+4]**

(b) Let  $T$  be a 2-tree with  $k$  leaves. Prove that the height  $h$  of  $T$  satisfies  $h \geq \lceil \lg k \rceil$  and the external path length  $E(T)$  satisfies  $E(T) \geq k \lg k$ . [5]

4. (a) Consider a binomial heap that comprises 27 nodes, and its minimum key value is 5 which is present in the largest binomial tree. Devise an algorithm that extracts the node with minimum key, and show how the algorithm works for the assumed binomial heap. [5]

(b) Devise a simple, easy-to-calculate hash function for mapping three- and four-letter words to integers between 0 and  $n-1$ , inclusive. Find the values of your function on the words: ALPA, MAP, BPUT, SET, TET, SAT, CTS, ALAP, PAM, SIT, DAC, JEE, BATA, TCS for  $n = 7$ . Try for as few collisions as possible. [5]

5. (a) Briefly state the *proof by local replacement* technique for proving a new problem NP-complete. Consider the following instance of SAT and convert it into a corresponding instance of 3SAT, thus proving 3SAT is also NP-complete.

$$\psi = (a' + b' + x + d' + e + h)(c + f + g')(c' + p + r' + s')$$

[2+3]

(b) Compute the smallest number of keys that will force a B-tree of order  $m$  to have three levels (assuming the root is at level one) when inserted in an appropriate order.

[5]

6. (a) Define sparse matrix and give an example. Discuss different storing mechanisms of sparse matrices in a computer along with their space complexities. [1+4]

(b) Suppose that  $L_1$  and  $L_2$  are two lists containing  $n_1$  and  $n_2$  integers, respectively, and both lists are already sorted in their numerical values in non-descending order. (i) Use the idea of binary search to describe how to find the median of the  $n_1+n_2$  integers in total without combining the lists. (ii) How could the same search technique help compute the median of the combined lists? [4+1]

7. During the Covid pandemic, the administration assigned beds to patients in different hospitals by guiding them to the correct hospital. A maximum number of patients get beds in a hospital. At any instant, the administration knows the number of patients,  $p$  who need beds in a hospital, and the number of free beds,  $b$  in each hospital. Design a flow network and describe the algorithm to help the administrator to assign beds to a maximum number of patients. State the time complexity of your algorithm with necessary justifications. [7+3]

8. (a) Determine the cost and structure of an optimal binary search tree for a set  $n = 7$  keys.  $p_i$ , the probabilities of occurrences of the key values,  $k_i$ ,  $1 \leq i \leq 7$  and  $q_i$ ,  $0 \leq i \leq 7$ , the probabilities of dummy keys,  $d_i$ , are given as follows:

| $i$   | 0    | 1    | 2    | 3    | 4     | 5     | 6     | 7     |
|-------|------|------|------|------|-------|-------|-------|-------|
| $p_i$ | –    | 0.02 | 0.03 | 0.04 | 0.01  | 0.05  | 0.06  | 0.07  |
| $q_i$ | 0.03 | 0.03 | 0.03 | 0.03 | 0.025 | 0.025 | 0.025 | 0.025 |

[7]

- (b) Give an algorithm to construct the longest common subsequence of two strings, X and Y, given the table that stores the respective indicators for the optimal solution. Compute the time complexity of this algorithm with justification. [3]